PAPER_694: Crab Nebula (M1/NGC 1952): SNR + Pulsar Wind Nebula UQFF

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Abstract

The Crab Nebula (M1/NGC 1952) is the remnant of SN 1054, containing a 33 ms pulsar driving a powerful pulsar wind nebula (PWN). UQFF modifications to the Sedov-Taylor expansion and synchrotron emission are derived.

1. Parameters

Parameter	Value
M_{ejecta}	$4.6M_\odot$
E_{SN}	10^{44} J
P_{pulsar}	33 ms
B_{pulsar}	3.8×10^8 T
Age	971 yr

2. Spin-Down Luminosity

$$L_{sd} = I|\Omega\dot{\Omega}|, \quad \dot{\Omega} = -\frac{B^2 R_{ns}^6 \Omega^3}{6Ic^3}$$

$L_{sd}(Crab) \approx 5 \times 10^{31}$ W.

3. UQFF SNR Expansion

$$v_{SNR,UQFF} = \sqrt{\frac{2E_{SN}(1-f_{TRZ})}{M_{ej}}}\left(1 + \frac{\rho_{SCm}}{\rho_{UA}}\right)$$

4. Synchrotron Cooling

$$t_{sync} = \frac{9m_e^3c^5}{4r_e^2 m_e c B^2 \gamma_e}$$

5. Sedov-Taylor Phase

$$R_{SNR}(t) = \left(\frac{E_{SN}}{\rho_0}\right)^{1/5} t^{2/5} \cdot \xi_0, \quad \xi_0 \approx 1.15$$

References

- Hester (2008), ARA&A, 46, 127
- UQFF Framework, Star-Magic Session 174

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